Exploring OPC UA Architecture and Digital Twins in Critical Infrastructures

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Abstract. This research seeks to demonstrate an initial exploration of the industry-standard open platform communication architecture for real-time data visualization of sensors and PLCs in critical infrastructures. The model critical infrastructures mentioned in this article are the WaterLAB, with its smart water-focused infrastructure, and the Lucas-Nuelle Savonia Smart Factory. We have examined different methodologies for configuring client-server architecture in data flow systems within selected critical infrastructures. According to our investigations, we have developed an architecture in which the WaterLAB's WaterLOOP distribution system simulator OPCUA server is initialized using the TwinCAT OPC UA configurator. For the client side, we conducted experiments to demonstrate real-time data on three distinct terminals: UaExpert, Prosys OPC UA Historian, and Prosys OPC UA Monitor. Furthermore, a prototype of a digital twin of an industrial control system has been developed, which incorporates programmable logic controllers (PLCs), sensors, and a communication network. This prototype has been implemented using the ignition platform. Additionally, we conducted an analysis of various OPC UA clients concerning their data security and data rate capabilities.

Keywords: Critical Infrastructure, OPC UA, PLC, Data Security, Digital Twin, Smart Water, Smart Factory

1 Introduction

In an era driven by digital transformation and Industry 4.0 [1], the integration of cutting-edge technologies has enabled remarkable advances in critical infrastructure. The OPC UA (Open Platform Communications Unified Architecture) [2] standard, together with its concept of a digital twin [3], is one of the key elements of this revolutionary journey. This preliminary study dives into the critical interaction between OPC UA architecture and Digital Twins, focusing on crucial infrastructures such as smart water plants and smart factories. The motivation for this research emerges from a vital need to improve the resilience, efficiency, and security of these systems.

Critical infrastructures serve as the foundation of modern society, providing valuable essentials and services from water supply and automated transportation to industrial

processes. The introduction of smart technologies has brought in an era in which these infrastructures are not only interconnected but also become more dependent on digital solutions for their optimum efficiency. The introduction of smart technologies has brought in an era in which these infrastructures are not only interconnected but also become more dependent on digital solutions for their optimum efficiency.

Understanding the OPC UA architecture and its application for digital twins can improve the reliability of critical infrastructure. This collaboration enables real-time data transmission, seamless communication across several devices, and the production of virtual models that replicate actual systems. This preliminary research on OPC UA architecture and Digital Twins in the context of critical infrastructures is an important step toward realizing the full potential of Industry 4.0 technology. By addressing the issues recognized by smart water plants and smart industries, our study seeks to set an example for a future in which our critical infrastructures are not only advanced technologically but also resilient in the face of growing threats and challenges.

The article is presented as follows. The initial section will describe motivation, critical infrastructures, and the OPC UA architecture. The sections that follow include discussions, future work, and a conclusion. This article has been expanded from the bachelor's degree thesis of the first author. [4]

2 Motivation and Critical Infrastructure

Integrating industry-standard open platform communication architecture has emerged as a major frontier in the never-ending quest for efficiency, resilience, and security in critical infrastructure [5]. The urgency and relevance of this research emerge from its enormous potential to improve the operating dynamics of these critical facilities.

This study undertakes an in-depth investigation of various configurations of client-server architecture designed for optimal data flow inside critical infrastructures. The TwinCAT OPC UA configurator is used to set up an OPC UA server in the smart WaterLOOP, as a standard example. The study continues to the client side, where real-time data visualization is demonstrated utilizing three different platforms: UaExpert, Prosys OPC UA Historian, and Prosys OPC UA Monitor.

One significant result of this effort is the achievement of a lightweight proof of concept for an industrial control system digital twin. This digital twin has PLCs, sensors, and a communication network, which was created using a prototype developed on the Ignition platform [6]. Aside from technological advancements, this study offers a detailed analysis of several OPC UA clients, evaluating their performance in terms of data security, network latency, and data rates.

As of 2022, the worldwide market value for cybersecurity solutions in critical infrastructure is expected to reach over $189 billion [7]. This is due to the growing need to protect key infrastructure such as electricity grids, transportation systems, and water treatment plants against cyber-attacks. Furthermore, the rising use of cloud-based technologies, the Internet of Things (IoT), and an increasing number of cyber-attacks on

critical infrastructure have all contributed to the rapid growth of the cybersecurity market in this region. The requirement for cybersecurity solutions in this industry is anticipated to rise rapidly in the upcoming years due to the rising reliance on technology and interconnection of critical infrastructure systems [8].

In summary, the article provides spotlight on a vital point in the continuous growth of smart water plants and smart factories. The next two subsections provide information on the facilities available at Savonia University of Applied Sciences

2.1 Savonia Smart WaterLAB

Savonia offers a specialized water research facility called WaterLAB, which contains WaterLOOP, a continuously operating Distribution System Simulator. WaterLOOP is an innovative testing platform that includes a one kilometer looped and branched pipeline network, water supply equipment, and primary and secondary pumping systems. It also includes with a duel waste water management system, which has a separate tank below the laboratory to separate non-dischargeable waste materials. The system can function in both open-loop and closed-loop modes, primarily using water provided by Kuopio. Its container-based treatment unit can produce potable water from a variety of sources. WaterLOOP comes with several high-precision online sensors that continuously monitor and record comprehensive water chemistry data in real time. When necessary, typical sampling and laboratory-grade analysis can be undertaken. The open pipeline architecture enables researchers to quickly add additional sensors or measuring devices as needed. This distribution system simulator allows for a variety of experimental scenarios, such as modeling critical processes and simulating contamination incidents. These might be a result of changes in water treatment practices, network maintenance activities, pipe blocks, flushing, cross-contamination, or even intentional harmful attacks on water supplies. Risks are reduced in this controlled and separated environment, giving operators complete control of system parameters. They can actively introduce physicochemical and microbiological contaminants, simulate hydraulic disturbances, or cause leak situations in a predictable manner. The system also allows for single- and multi-parameter alterations, including short-term, medium-term, and long-term changes in water quality. Various combinations of influencing elements can also be established.

WaterLOOP offers both random water quality changes and planned experimental studies. This feature enables the integration of real-time monitoring and simulation to develop tools like as decision support systems, situational awareness platforms, anomaly detection methods, and the testing of new technologies or services. The data produced by WaterLOOP can be utilized to train and test artificial intelligence models. WaterLOOP architecture includes a commercial-grade SCADA system and customizable PLC configuration, simulates real-world water distribution systems. This makes it especially useful for cybersecurity applications that include vulnerability assessments and penetration testing, and for simulating and analyzing critical operational scenarios.

In WaterLAB premises, Savonia operates private 5G New Radio Millimeter Wave (NR mmWave) with frequencies above 24 GHz. Available bands are n257 (indoors only) and n258 (indoors and outdoors). For NB-IoT test networks, available bands are B31,

B20, and B3 (450 MHz – 1800 MHz). Both these wireless networks are terminated to the same server, enabling data aggregation from multiple sources. [9]

2.2 Savonia Smart Factory

The Internet of Things (IoT) section at Savonia University of Applied Sciences has been developing modern education to promote smart factory development in the context of the Fourth Industrial Revolution (Industry 4.0). Under the Auto-TEK regional EU-funded project, we have developed the Lucas Nuelle smart factory environment and implemented several use cases, such as sensor data transfer over 5G private networks for critical, high data rate, and low latency applications. The smart factory is covered by Savonia's 5G NR SA (sub-6GHz) private network (band n77). In addition, we have also integrated an autonomous machine vision-assisted platform to check the production quality in real time. Furthermore, in our recent Virtual Learning Environment for Smart Factory (VLEFACT) funded by Erasmus plus, four European schools from Denmark (Mercantec), Germany (Carls Bens Schulle), and Finland (Savonia, and YSAO) have been creating a novel learning environment for smart factory education. Savonia University of Applied Sciences is responsible for 1) Factory data collection, analysis, and visualization, and 2) data-driven manufacturing. [10]

3 OPC UA Server Client Architecture

We have implemented OPC UA server-client architecture, as illustrated in Figure 1. The OPC UA architecture configuration encompasses communication protocols, multiple client applications for data processing, and a visualization platform for data analysis and presentation.

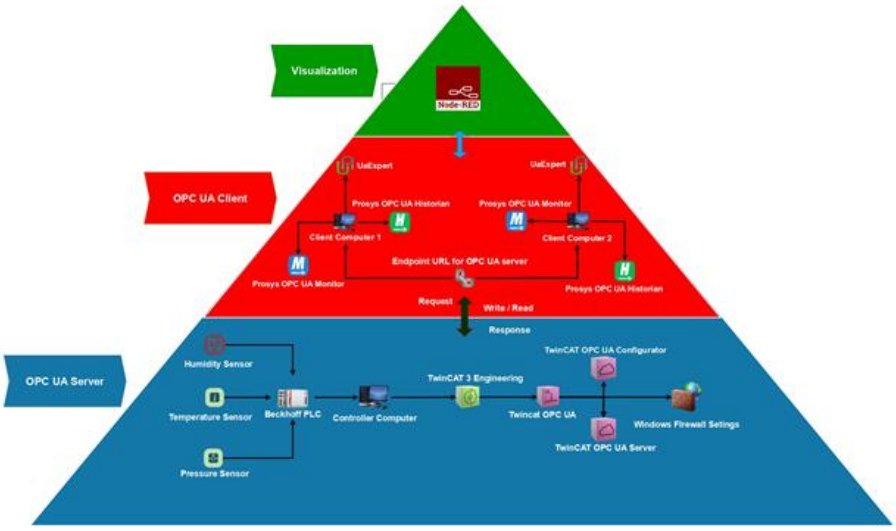

Figure 1. Implemented OPC UA Server-Client Architecture for WaterLOOP

The OPC UA Server is the core of the system architecture which facilitates communication between various sensors connected to the Beckhoff CPU module and the other components of the system. The PLC system integration with a server/controller computer that runs TwinCAT 3 Engineering (TE1000). In this scenario, the TwinCAT OPC UA software is used to deploy and configure the OPC UA server [11]. The TwinCAT OPC UA Configurator is mainly responsible for server configuration, and the TwinCAT OPC UA Server functions as the runtime component. To ensure seamless communication between the server and client devices on the local network, the correct firewall rules must be set up to allow inbound traffic on the selected OPC UA port. In this setup, the server computer acts as a communication hub, allowing for real-time data transmission with read and write features. The client sends requests to the server and receives responses, ensuring that data flows consistently and efficiently across the system.

UaExpert and Prosys are client applications utilized during the OPC UA client phase to establish a connection with an OPC server. UaExpert functions as a testing client and offers capabilities such as data access, alarms and conditions, historical access, and the activation of UA methods. The Prosys OPC UA monitor facilitates the creation of a dashboard shown in Figure 2 that emphasizes the display of measurement data and the live monitoring of production processes and sensors linked to the OPC UA server. In addition, Prosys Historian gathers live data and stores it in SQL databases for archiving and analysis. The architecture incorporates Node-RED, a platform that retrieves OPC UA data from the server and presents it in an understandable format on a Node-RED Dashboard for data visualization. [4]

3.1 UaExpert and Prosys OPC UA

UaExpert, an advanced OPC UA client, was essential in connecting with the WaterLOOP OPC UA server, and included applications includes DataAccess, Alarms and Conditions, Historical Access, and UA Method calls. It significantly raised the efficiency of server contact, decreased latency, and streamlined data collecting. UaExpert's essential design includes inspecting certificates, detecting OPC UA servers, connecting servers and clients, investigating the information model, and displaying UA node properties and references [12]. More effective access to and use of the WaterLOOP OPC UA server was made possible by the Address Space pane and Attribute window, which improved understanding of the information model of the server. Effective system monitoring and analysis were supported by the Data Access View, which offered real-time insights into node statuses, values, and timestamps from many sensors. The application showed consistent performance when it attempted modifying sensor data in the server environment by performing write operations on certain nodes. Exploring additional functionalities such as method calls and history access may help to increase the WaterLOOP OPC UA Server's capabilities.[4]

Another highly functional and adaptable OPC UA client that is connected to the WaterLOOP OPC UA server is called Prosys OPC UA Monitor [13]. A customized dashboard was created to monitor sensor data from the WaterLOOP PLC, including a

variety of visual gauges and graphical components. Each item was carefully created with optimal features to provide efficient and user-friendly connectivity with the system.

Figure 2. Prosys UPC-UA Monitor

The volume of data transmission and rapid visualization can be controlled by setting the sample interval, which allows for quick responses to changes in data sources. It is simple to set up the measurement of value-based notifications and their visualization. The integrated data logger is easily activated, and it offers real-time gauge monitoring as well. This application provides a wide range of useful information that can be simply exported and used in other software for further research. After connecting to the WaterLOOP OPC UA server, node data can be exported to a CSV file with little effort. When working with large amounts of data, the ability to control the output using a variety of filters and other parameters is quite useful. The generated file might be processed for additional purposes or used to document. [4]

The Prosys OPC UA Historian and the WaterLOOP OPC UA server both were integrated to provide a dependable way to gather and document important data from automation systems with suitable sensors. Prosys OPC UA Historian is an excellent platform for collecting sensor data in real time and effectively storing it in SQL databases [14]. To ensure operational safety, Prosys OPC UA Historian integrates easily into current WaterLOOP automation frameworks and serves as a competent data logger that is compatible with SQL databases. Because of this connectivity, we have complete control over data management and can access important datasets at any time. Having all data behind a single point of access allows for more effective firewall rules and network administration. This can greatly improve security while simplifying our system design. Significantly, Prosys OPC UA Historian builds the foundation for optimal data usage

in the WaterLOOP OPC UA server environment and emerges as a key building block in the development of complete systems, beginning with sensor data gathering [4].

3.2 Node-Red

The Node-RED Dashboard [15] was used as a customized OPC UA client for the WaterLOOP OPC UA Server to improve sensor data visualization and system control. The dashboard not only visualizes real-time data, but it also provides a better understanding of the OPC UA server's complicated processes by efficiently gathering and displaying important variables such as temperature, pressure, and flow values via comprehensible gauges and charts.

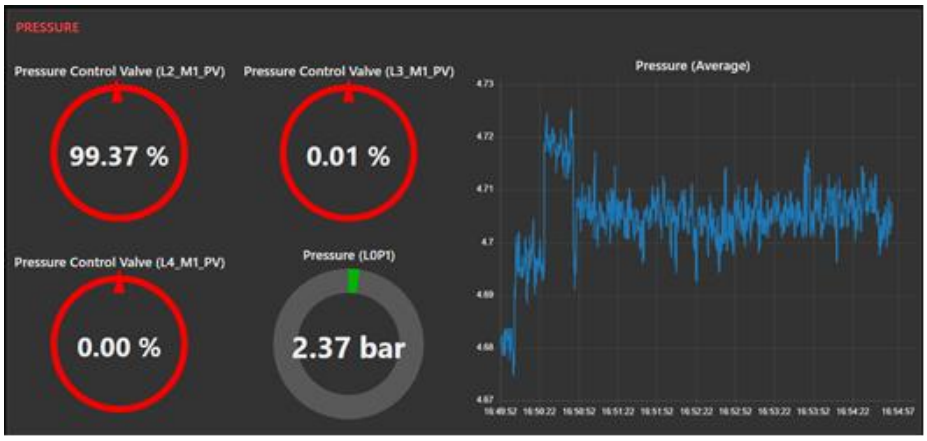

Figure 3. Smart WaterLAB Pressure Data Visualization

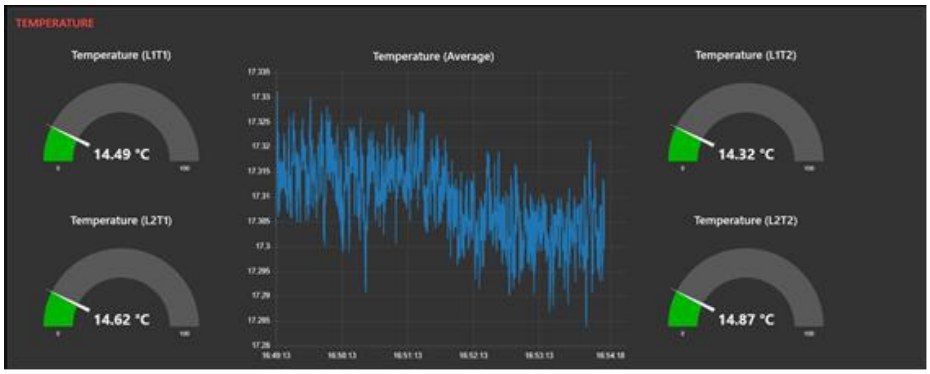

Figure 4. Smart WaterLAB Temperature Data Visualization

The dashboard accomplished various objectives, including presenting an overview of the system's structure and allowing users to navigate the hierarchical server address space structure with ease. It also highlighted the complexity and diversity of the data

collected by showcasing the flexibility of data categories and visualization approaches. The dashboard also enabled two-way data engagement, expanding its usefulness beyond typical data presentation. This feature improved operational management and real-time data processing by allowing users to write OPC UA values directly to the server via the dashboard. A full picture of system conditions was also made possible by the integration of events and alarms from the OPC UA server, allowing for quick response and proactive management. Figures 3 and 4 show how pressure and temperature data from the water laboratory were effectively displayed using Node-RED technology. [4]

4 Smart Factory Digital Twin Implementation

There are many remaining queries about the smart manufacturing facility's digital twin. The key problems are estimating the factory's production rate, evaluating the allocation of equipment and human resources during real-time production, and measuring the speed or number of items produced in each timeframe. To answer these questions, we used a 3D visual components tool [16], where items were first created and then customized with the appropriate properties from the library. Figure 6 demonstrates the real-time effectiveness of humans and automated machines. SolidWorks was used to model the product parts. Furthermore, we observe a precise measurement of the distance travelled by the individual (particularly Anna) in millimeters during a specified period, as well as a real-time picture of how she utilizes her time. Two feeder lines are available: one for the white product and another for the black product. In the upper right corner is a visual representation of the time allocation. In contrast, in the lower right corner is a numerical display of the quantity of items transported. There is a total of seven peaks; therefore, a total of seven parts were transported.

Figure 5. Savonia Smart Factory

Figure 5 illustrates the physical Savonia Smart Factory, which is equipped with one master PLC and eight slave PLCs integrated to operate cohesively within an environment comprising Enterprise Resource Planning (ERP) and Manufacturing Execution System (MES) platforms. Communication within the system is facilitated through ProfiNet and the Constrained Application Protocol (CoAP).

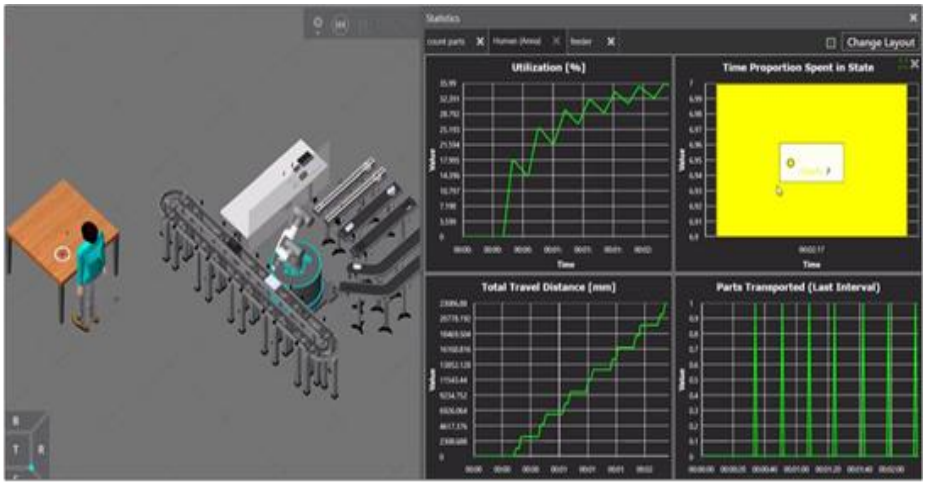

Figure 6. Digital Twin Implementation Using 3D Visual Components

5 Water Plant Digital Twin

The development of a modern digital twin model utilizing Ignition HMI software marks an important step in the research of WaterLOOP infrastructure control and monitoring. This innovative tool provides a comprehensive and continuous visualization of the critical parameters by seamlessly integrating real-time sensor data from five different sensors in WaterLOOP. The architecture is improved by including an Ignition OPC UA server within the server computer, which enables the client computer with the digital twin mechanism to get the most recent data fast and securely. This technology allows users to run the water tanks and pumps in real-time while also enabling continuous monitoring of important data.

Using Ignition HMI software ensures a user interface that is simple to operate and renders the system easier to access and control [17]. As an important accomplishment in the field of smart infrastructure, this technological ability not only illustrates the possibilities of digital twin applications in industries but also sets a foundation for smart and adaptive water management systems.

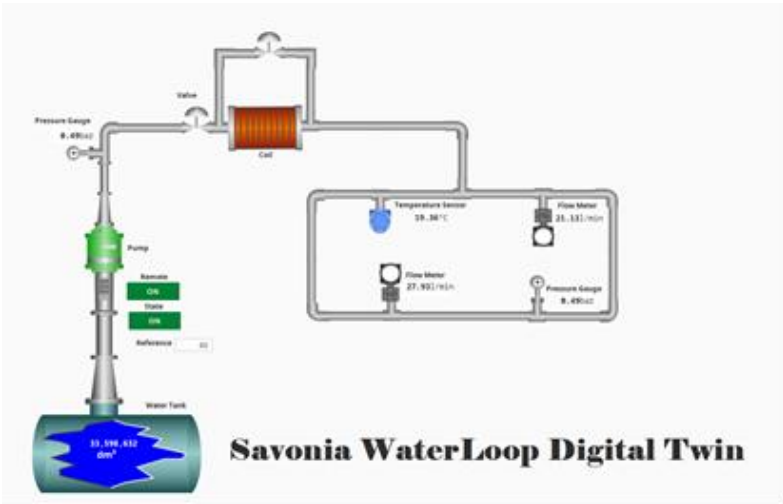

Figure 7. Savonia WaterLOOP Digital Twin [4]

6 Comparative Analysis of OPC UA Clients

Table 1 shows the comparative analysis of the data security versus data transmission rate in different visualization scenarios. In this analysis, typical OPC UA clients such as UaExpert, Prosys OPC UA Monitor, and our own self-developed client are compared and analyzed.

Table 1: Comparative Analysis of different scenarios [18]

OPC UA Clients	Data Security	Data Rate
UaExpert	Features like encryption and authentication are included in UaExpert to enhance data security. To enable safe client-server communication, it aligns with the different OPC UA security requirements. Role-based access control, encrypted communication, and user authentication all enhance data security.	Factors including network latency, server response times, and the client's data processing ability have an impact on total data rate performance.
Prosys OPC UA Monitor	The Prosys OPC UA Monitor offers enhanced data security features. To guarantee connection security, the	Prosys OPC UA Monitor can handle various data speeds. The client's capacity to connect

	client supports encryption protocols and authentication techniques. Fine-grained control over data is made possible via certificate-based authentication and access control.	to OPC UA servers and accept incoming data would decide how well it performed.
Self-Built Client	The developers who created a customized client are fully responsible for its security. Three essential security techniques to guarantee data protection are encryption, authentication, and authorization. Developers must adhere to OPC UA security specifications and best security practices to design a safe client.	The data rate performance of a self-developed client is mainly determined by how well developers control incoming data and optimize their code. To verify that the client can manage the desired data rate without delay, extensive performance testing is required.

7 Future Work and Conclusion

The work depicted in this article is currently undergoing development. Currently, we have the capability to display the sensor's data in real-time on multiple dashboards. While some research is still ongoing, we have successfully developed a proof of concept to demonstrate the visualization of real-time data in our Savonia WaterLAB. We are presently utilizing MATLAB Simulink and Visual Components to implement water plant and smart factory systems. These systems will collect data and display it in real-time, ultimately improving the efficiency and productivity of the plants. Simultaneously, we are presently engaged in utilizing Siemens NX software to construct three-dimensional models of our facilities.

In summary, this study shows notable advances in continuous data visualization for sensors and PLCs in critical infrastructure systems. It highlights the usage of highly adopted open-platform communication architectures, which are frequently used in industrial environments.

This research accomplishes two main objectives. Firstly, it establishes a solid basis for real-time data visualization by effectively managing the intricacies of configuring client-server architecture in data flow systems. Secondly, it addresses crucial factors that need to be considered to guarantee the security and efficiency of these systems in critical infrastructures. This work establishes the foundation for future progress in incorporating open platform communication architectures in industrial environments, promoting innovation and dependability in smart water and factory technologies.

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